

SERGIO DANIEL HERNANDEZ CHARPAK

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French ◦ Colombian ◦ C Swiss Permit

Research Overview

I am a passionate researcher on the interface between artificial intelligence / machine learning in healthcare, focusing on leveraging and integrating neuroimaging strategies with implantable neuroprosthetics to restore neurological functions in individuals with spinal cord injury, Parkinson's disease and other neurological conditions.

Education

École Polytechnique Fédérale de Lausanne

Electrical Engineering, PhD Candidate

Neuroimaging to Guide the Delivery of Implantable Neuroprosthetics

Restoring Neurological Function

.NeuroRestore

Lausanne, Switzerland

September 2020-February 2025

École Polytechnique Fédérale de Lausanne

Computational Science and Engineering, Master Thesis Student

Lausanne, Switzerland

September 2017-February 2020

Universidad de los Andes

Physics, Bachelor of Science

Computing Engineering, Bachelor of Engineering

Japanese Language and Culture, Minor

Bogotá, Colombia

January 2010 - March 2017 - GPA **4.23**/5.00

January 2010 - March 2017 - GPA **4.23**/5.00

Kyoto Institute of Culture and Language

Intermediate Japanese Student

Kyoto, Japan

October 2013-March 2014

Work Experience

EPFL

*.NeuroRestore - Defitech Center
for Interventional Neurotherapies*

Lausanne, Switzerland

March 2025 - Present

Imaging and Computation Manager – .NeuroRestore - Clinical Division

Setting up the medical imaging, computational modelling and simulation platform for guidance of implantable neuroprosthetics to restore neurological function.
Supervision of PhD student Philippe Forero.

EPFL

*.NeuroRestore - Defitech Center
for Interventional Neurotherapies*

Lausanne, Switzerland

September 2020 – February 2025

Doctoral Assistant – .NeuroRestore – Clinical Division

Neuroimaging to guide the delivery of implantable neuroprosthetics restoring neurological function. Automatization of personalized 3D models for simulations of spinal cord stimulation paradigms for patients suffering spinal cord injury or other neurological dysfunctions under the supervision of prof. G. Courtine and prof. J. Bloch

EPFL

G-Lab

Geneva, Switzerland

February - September 2020

Scientific Assistant – Neurorestore – Computational Neuroscience Unit

Contributing with image processing, data analysis in several research projects around personalized targeted spinal cord stimulation paradigms for spinal cord injury patients.

Nagra Kudelski Group

Cloud

Cheseaux, Switzerland

February – July 2019

Internship – Cloud Infra Team

Enabled real-time deep learning in production for anomaly detection in data streams using Pytorch, TF, Scala, Spark and Deeplearning4J. Developed neural networks models for unsupervised anomaly detection on time series deploying them for real time alert generation under supervision of Arnaud Gaillard.

Research Experience

Curriculum Vitae

EPFL

G-Lab & MIP Lab

Geneva, Switzerland

September 2019 - February 2020

EPFL

G-Lab

Geneva, Switzerland

August 2018 – February 2019

EPFL

MIP Lab

Geneva, Switzerland

February - July 2018

Universidad de los Andes

Department of Computing

Engineering

Bogotá, Colombia

August - December 2016

Laboratoire CPPM

LSST Project

Marseille, France

June 2016

Universidad de los Andes

Department of Physics

Bogotá, Colombia

January - May 2016

Universidad de los Andes

School of Engineering

Bogotá, Colombia

August 2015 - December 2016

Fermi National Laboratory

Neutrino Division

Batavia, U.S.A.

June 2015 – August 2015

Tokyo University of Marine

Science and Technology

Tokyo, Japan

May 2024 - June 2014

Master Thesis - Prof. Courtine's Laboratory & MIP Laboratory (prof. Van De Ville)

Conducted a lumbar Spinal Cord (SC) fMRI study to deconstruct segmental innervation of sensorimotor circuits in the lumbosacral SC in healthy. Integrated it into personalized targeted SC stimulation paradigms for patients under the direction of PhD students A. Rowald, N. Kinany, prof. G. Courtine and prof. D. Van De Ville.

Master Semester Project - Prof. Courtine's Laboratory

Artificially represented brain input to spinal sensorimotor circuits through the implementation of a DL framework for unsupervised and supervised learning strategies to drive a biomechanical model of the lower limbs in human under the direction of PhD student A. Rowald and prof. Gregoire Courtine.

Master Semester Project - Medical Image Processing Laboratory

Processed high resolution 7-Tesla brain fMRI data fMRI data using the Total Activation method, and found the innovation-driven Co-Activation Patterns (iCAPs) and their time behaviors on three different paradigms. Worked under the direction of PhD student A. Tarun and prof. Dimitri Van De Ville.

Undergraduate Thesis

Implemented part of an Image Analysis tool for the Segmentation of the aorta artery for quantifications of the elasticity of the aorta artery and calcifications under the direction of prof. Marcela Hernandez and prof. Leonardo Florez.

Internship - LSST Project

Studied and implemented different image processing and statistic techniques for the detection of transients in astrophysical images. Under the supervision of scientist Dominique Fouchez.

Undergraduate Thesis

Titled Laniakea in a Cosmological Context. Detected galaxies superclusters in simulated cosmological structures based on galaxies velocities properties under the direction of prof. Jaime E. Forero.

Undergraduate Research Assistant

Developed Python tools for testing prototypes in the project Astronomical Image processing from large all-sky photometric surveys for the detection and measurements of transients under the mentorship of prof. Marcela Hernandez.

IPM Intern – Muon G-2 Experiment

Part of the team for the Test Beam of a Straw Detector Prototype in charge of the High Voltage and assisted with the analysis of the data taken under the mentorship of scientist Brendan C Casey.

Visiting Student - Control and Robotics Laboratory

Assisted with the integration and control of a helicopter with Arduino under the supervision of professors Sho and Ito.

Teaching Experience

EPFL

Lausanne, Switzerland
2020,2021,2022,2023,2024,
2025,2026

Master Thesis & Internships Supervisors

- Millisecond-aligned muscle activity (EMG), epidural electrical stimulation (EES) configurations and marker-less kinematics for data-driven strategies, Viva Berlinghi (EPFL)
- Biophysically informed hybrid model for Upper-Limb EMG Prediction in Spinal Cord Injury EES therapies, Gonçalo Braga (EPFL)
- Medical imaging data platform for training AI models, Noé Pratt (EPFL)
- Inter-individual training strategies for muscle responses prediction from spinal cord stimulation, Constance de Trogo (EPFL)
- Self-supervised learning of human spinal cord anatomy - SpineGPT, Daniel Selmin (EPFL)
- Finding data-driven robust functional and structural spinal levels, Olivia Ruggaber (EPFL)
- Quality enhancement of MRI volumes using with adversarial training with masked autoencoders, Bilel El-Guallali (EPFL).
- Proprioceptive map of the human subcortical neural motor network, Meriem Bourouba (Université Paris Cité)
- Recurrent neural networks to study and control motor execution, Marion Pavaux (Université Paris-Saclay, École des Mines de Saint-Étienne)
- Data-driven identification of lower limbs proprioceptive projections mapping, Neala Rohner (ETHZ)
- M3D-SAM: Multi-purpose multi-modal zero-shot semi-to-fully automatic segmentation on medical imaging based on SAM, Hanane Moha-Ouchane (TUM)
- Home-monitoring of gait for Parkinson's disease patients, Maxime Pillet (EPFL)
- Automated Gait Analysis to Characterise Locomotor Deficits in Home Environments for Parkinson's Patients, Ian Enderli (EPFL)
- Semi-automatic extraction of the spinal root centerlines, Raphaël Mariétan (EPFL)

EPFL

Lausanne, Switzerland
2020,2021,2022,2023,2024,
2025

Semester Master projects

- Personalized bioelectrical model reconstruction for transcutaneous Spinal Cord Stimulation (TSCS), Luis Fennessy (EPFL)
- Incorporating expert semantic annotations into a large spinal cord MRI dataset, Wenxin Che (EPFL)
- Robustification of Personalized Spinal Cord Model Reconstruction, Chléa Schiff (EPFL)
- Learning the Natural Language of the Spinal Cord using Masked Autoencoders, Léon Muller (EPFL)
- Learning the natural language of the spinal cord, Juliette Hars (EPFL)
- Robustification of AI pipeline for personalized model reconstruction, Nicolas Reategui (EPFL)
- Self-supervised for MRI artifact detection and reconstruction, Jonas Blanc (EPFL)
- Automatic, accurate and robust vertebrae-based CT to MRI co-registration, Benkirane, Ismail (EPFL)
- Artificial biologically relevant spinal networks linking electrical stimulation and muscle responses in time, Axelle Piguet (EPFL)
- Segmentation and Identification of Vertebrae in CT scans, Aline Brunner (Bachelor - EPFL)
- Deep learning network for the segmentation of spinal roots, Quentin Uhl (EPFL)
- A computational approach to study control and execution of locomotion, Daniele Hamm (EPFL)

EPFL

Lausanne, Switzerland
2021,2022,2023,2024

Graduate Teaching Assistant

Teaching Assistant for Sensorimotor Neuroprosthetics, Assistive Technology Challenge courses. Supervised group of students in the development of assistive technologies to solve specific challenges from individuals with disabilities.

Universidad de los Andes Undergraduate Teaching Assistant

Bogotá, Colombia

2011,2012,2013,2014,2015

Teaching Assistant for Object Oriented Programming, Data Structures, Modeling, Simulation and Optimization, and Computational Methods courses.

Publications

1. A digital bridge to improve arm and hand movements after tetraplegia, Collin, T., Yu I.B., Sakr, I., **Hernandez-Charpak, S. D.**, et. al., **Nature Medicine** (In revision)
2. Learning the natural language of the human brain with BrainGPT, Sakr, I., Spagnolo, V., Collin, T., Dahan S., Van Delft B., Martel F., Yu I.B., Coquoz L., **Hernandez-Charpak, S.D.**, et al., **Science** (In revision)
3. Schlienger, R., Landelle, C., **Hernandez-Charpak, S.D.**, Pinzon-Corredor, D.M., Caron-Guyon, J., Sein, J., Nazarian, B., Bloch, J., Felician, O., Anton, J., Courtine, G., Kavounoudias, A., 2025. Mapping Human Proprioceptive Projections of Upper Limb Muscles Through Spinal Cord fMRI. **Human Brain Mapping** 46, e70386. <https://doi.org/10.1002/hbm.70386>
4. Hankov, N., Caban, M., Demesmaeker, R., Roulet, M., Komi, S., Xiloyannis, M., Gehrig, A., Varescon, C., Spiess, M. R., Maggioni, S., Basla, C., Koginov, G., Haufe, F., D’Ercole, M., Harte, C., **Hernandez-Charpak, S. D.**, et al. (2025). Augmenting rehabilitation robotics with spinal cord neuromodulation: A proof of concept. **Science Robotics**, 10(100), (2025). <https://doi.org/10.1126/scirobotics.adn5564>
5. **Hernandez-Charpak, S. D.**, Kinany, N., Ricchi, I., Schlienger, R., Mattera, L., Martuzzi, R., Nazarian, B., Demesmaeker, R., Rowald, A., Kavounoudias, A., Bloch, J., Courtine, G., & Van De Ville, D. (2025). Towards personalized mapping through lumbosacral spinal cord task fMRI. **Imaging Neuroscience**, 3, imag_a_00455, (2025), https://doi.org/10.1162/imag_a_00455
6. Phillips, A.A., Gandhi, A.P., Hankov, N., **Hernandez-Charpak, S.D.**, et al., 2025. An implantable system to restore hemodynamic stability after spinal cord injury. **Nature Medicine** 31, 2946–2957. <https://doi.org/10.1038/s41591-025-03614-w>
7. Milekovic, T., Moraud, E.M., Macellari, N., Moerman, C., Raschellà, F., Sun, S., Perich, M.G., Varescon, C., Demesmaeker, R., Bruel, A., Bole-Feysot, L.N., Schiavone, G., Pirondini, E., YunLong, C., Hao, L., Galvez, A., **Hernandez-Charpak, S.D.**, et al, 2023. A spinal cord neuroprosthesis for locomotor deficits due to Parkinson’s disease. **Nature Medicine** 29, 2854–2865. <https://doi.org/10.1038/s41591-023-02584-1>
8. Lorach, H., Galvez, A., Spagnolo, V., Martel, F., Karakas, S., Interling, N., Vat, M., Faivre, O., Harte, C., Komi, S., Ravier, J., Collin, T., Coquoz, L., Sakr, I., Baaklini, E., **Hernandez-Charpak, S. D.**, et al. (2023). Walking naturally after spinal cord injury using a brain–spine interface. **Nature**, 618(7963), 126–133. <https://doi.org/10.1038/s41586-023-06094-5>
9. Squair, J. W., Berney M., Castro Jimenez M., Hankov N., Demesmaeker R., Amir S., Paley A., **Hernandez-Charpak S.D.**, et al. “Implanted System for Orthostatic Hypotension in Multiple-System Atrophy.” **New England Journal of Medicine** 386, no. 14 (April 7, 2022): 1339–44. <https://doi.org/10.1056/NEJMoa2112809>.
10. Rowald, A., Komi, S., Demesmaeker, R., Baaklini E., **Hernandez-Charpak S.D.**, et al. Activity-dependent spinal cord neuromodulation rapidly restores trunk and leg motor functions after complete paralysis. **Nature Medicine** 28, 260–271 (2022). <https://doi.org/10.1038/s41591-021-01663-5>
11. Peñaranda-Rivera J. D., Paipa-León D. L., **Hernández-Charpak S.D.**, Forero-Romero J.E., Superclusters from velocity divergence fields, **Monthly Notices of the Royal Astronomical Society: Letters**, Volume 500, Issue 1, January 2021, Pages L32–L36, <https://doi.org/10.1093/mnras/laa177>

Awards

BCI Awards

*Vienna, Austria
October 2025*

3rd place winner

An implantable brain-spine interface restoring lower limb movements after complete spinal cord injury, Spagnolo, V., Sakr, I., Coquoz, L., Dahan, S., Carparelli, G., Abranches, P., Interling, N., Deschenaux, C., **Hernandez-Charpak, S.D.**, et al.

BCI Awards

*Vienna, Austria
October 2024*

2nd place winner

A brain to spinal cord digital bridge to restore voluntary arm and hand movements after a spinal cord injury, Collin, T., Sakr, I., Spagnolo, V., Interling, N., Hervé, J., **Hernandez-Charpak, S.D.**, et al..

BCI Awards

*Vienna, Austria
October 2022*

1st place winner

A brain to spinal cord digital bridge to restore voluntary arm and hand movements after a spinal cord injury, Lorach, H., Galvez, A., Spagnolo, V., Martel, F., Karakas, S., Interling, N., Vat, M., Faivre, O., Harte, C., Komi, S., Ravier, J., Collin, T., Coquoz, L., Sakr, I., Baaklini, E., **Hernandez-Charpak, S.D.**, et al..

**From Technology
to Treatment
Advancing Precision
Medicine**

*Zurich, Switzerland
August 2025*

Poster – Best pitch & poster award

Digital twin framework for personalized epidural electrical stimulation in spinal cord injury therapies, Forero P. **Hernandez-Charpak S.D.**, et al.

Neuro-X Conference

*Geneva, Switzerland
December 2022*

Oral talk – Best presentation award

Beyond the brain. Disentangling spinal cord function using fMRI, Kinany N., Ricchi I., **Hernandez-Charpak S.D.**.

Conferences

**Organization for Human
Brain Mapping**

*Bordeaux, France
June 2026*

- SpineGPT: Self-supervised learning of human spinal cord anatomy for clinical downstream tasks, **Hernandez-Charpak S.D.**, Sakr I., Selmin D., Muller L., Hars J., Blanc J., El-Ghallali B., Forero P., Ledoux J.B., Becce F., Bloch J., Courtine G., Lorach H.
- Potential of Personalized Spinal Cord Dynamic Functional Connectivity Across Regions and Pathologies, Ruggaber O., **Hernandez-Charpak S.D.**, Ricchi I., Ledoux J.B., Becce F., Bloch J., Martin-Moraud E., Courtine G., Lorach H., Van De Ville D.
- Spinal functional MRI to Map Proprioceptive Circuits in Cervical and Lumbar Regions, Schlienger R., Landelle C., Pinzon-Corredor D., **Hernandez-Charpak S.D.**, Théfenne L., Nazarian B., Sein J., Courtine G., Anton J.L., Kavounoudias A..

Society for Neuroscience

*San Diego, CA, USA
November 2025*

Posters

- Purpose-built spinal cord neuroprosthesis alleviates gait deficits in individuals with advanced Parkinson's disease, Varescon C., Sanchez Lopez P., Mauguin E., Deschenaux C., Bole-Feysot L., **Hernandez-Charpak S.D.**, et al.
- Closed-Loop Epidural Electrical Stimulation of the Lumbosacral Spinal Cord to Address Gait Impairments and Fluctuations in Advanced Parkinson's Disease, Sanchez Lopez P., Varescon C., Mauguin E., Deschenaux C., Bole-Feysot L., **Hernandez-Charpak S.D.**, et al.
- A digital bridge between the brain and spinal cord that restores lower limb movements after complete spinal cord injury, Spagnolo, V., Coquoz, L., Carparelli, G., Interling, N., Deschenaux, C., **Hernandez-Charpak, S.D.**, et al.

**European Society for
Magnetic Resonance in
Medicine and Biology**

*Marseille, France
October 2025*

**From Technology
to Treatment
Advancing Precision
Medicine**

*Zurich, Switzerland
August 2025*

BCI society Meeting

*Banff, AB, Canada
June 2025*

Rehab Week

*Chicago, IL, USA
May 2025*

Society for Neuroscience

*Chicago, IL, USA
October 2024*

Society for Neuroscience

*Washington D.C., USA
November 2023*

- Restoring Hemodynamic stability with spinal cord stimulation in Parkinsonian syndromes, Gandhi A.P. Amir S., Soriano J.E., Hudelle E., Hankov N., Nikolaeva A., **Hernandez-Charpak, S.D.**, et al.
- A digital bridge between the brain and spinal cord that restores arm and hand movement after spinal cord injury, Yu I.B., Collin T., Sakr I., Spagnolo V., Hernandez-Charpak S.D., et al.
- Long term use of a digital bridge to restore lower limb function, Coquoz, L., Spagnolo, V., Interling, N., Deschenaux, C., Galvez A., Carparelli, G., **Hernandez-Charpak S.D.**, et al.

Poster

Self-supervised learning of human spinal cord anatomy - SpineGPT, **Hernandez-Charpak S.D.**, Sakr I. et al.

Poster – Best poster award

Digital twin framework for personalized epidural electrical stimulation in spinal cord injury therapies, Forero P. **Hernandez-Charpak S.D.**, et al.

Posters

- An implantable brain-spine interface restoring lower limb movements in a patient with motor complete spinal cord injury, Spagnolo V., Carparelli G., Coquoz L., Deschenaux C., **Hernandez-Charpak S.D.**, et al.
- From ECoG signals to cervical epidural electrical stimulation restoring arm & hand movements after a spinal cord injury, Collin T., Yu I.B., Spagnolo V., Sakr I., **Hernandez-Charpak S.D.**,

Posters

- Reconnecting mind and movement: walking rehabilitation in chronic spinal cord injury enhanced with a brain-spine interface, Interling N., Spagnolo V., Galvez A., Carparelli G., Deschenaux C., Hankov N., Coquoz L., Abranches P., Martel F., Karakas S., Vat M., Faivre O., Harte C., Sasportes C.D., Simondin M., Komi S., Ravier J., Collin T., Sakr I., Baaklini E., **Hernandez-Charpak S.D.**, et al.
- ARCIM Therapy: A new targeted spinal cord neuromodulation therapy to enhance mobility in individuals with advanced Parkinson's Disease, Deschenaux C., Sanchez López P., Mauguin E., Varescon C., Bole-Feysot L., **Hernandez-Charpak S.D.**, et al.
- Augmenting rehabilitation robotics with spinal cord neuromodulation a proof of concept, Hankov N., Caban M., Demesmaeker R., Roulet M., Komi S., Xiloyannis M., Gehrig A., Varescon C., Spiess M.R., Maggioni S., Basla C., Koginov G., Haufe F., D'Ercole M., Harte C., **Hernandez-Charpak S.D.**, et al.

Posters

- Interfacing brain-decoded motor intentions with the cervical spinal cord to restore voluntary arm and hand movements, Collin T., Spagnolo V., Sakr I., Carparelli G., **Hernandez-Charpak S.D.**, et al.

Posters

- Automated Generation of Spinal Cord Models for Pre-Operative Planning of Epidural Electrical Stimulation Surgical Interventions, **Hernandez-Charpak S.D.**, et al.
- Non-Invasive And Invasive Methodologies of Stimulating the Cervical Spinal Cord in Order to Improve Upper-Limb Functions After

Tetraplegia, Collin T., Spagnolo V., Berny M., Interling N., Hervé J., Sakr I., Galvez A., Carparelli G., **Hernandez-Charpak S.D.**, et. al.

- *Automated Configuration of Epidural Electrical Spinal Cord Stimulation for Neurological Disorders*, Carparelli G., Abranches P., Macellari N., Komi S., Yang X., Harte C., Dumont G., Collin T., **Hernandez-Charpak S.D.**, et. al.

Society for Neuroscience

San Diego, CA, USA
November 2022

Posters

- *A Spinal Cord Neuroprosthesis That Alleviates Locomotor Deficits In PD*, Milekovic T., Moraud E.M., Nicolo Macellari, Moerman C., Raschellà F., Sun S, Perich M.G., Varescon C., Demesmaeker R., Bruel A., Bole-Feysot L.N., Schiavone G., Jianzhong Y., Galvez A., **Hernandez-Charpak S.D.**, et. al.
- *A Task-Adaptive Spinal Neuroprosthesis to Improve Gait Deficits in People with Parkinson's Disease*, Macellari N., Moerman C., Sasportes C.D., Harte C., Varescon C., Bole-Feysot L.N., Carparelli G., **Hernandez-Charpak S.D.**, et. al.

XV LARIM (Latin

American Regional IAU Meeting)

Cartagena, Colombia
October 2016

Oral Talk

- **Laniakea in a Cosmological Context:** Galaxies superclusters detection in simulated cosmological structures under the direction of prof. Jaime E. Forero.

Skills

Software and programming

Proficient

Python, MATLAB, C, C++, Java, Git, LaTeX,
Bash, Pytorch, Microsoft Powerpoint, Adobe
Illustrator

OS: Linux, Windows and Mac OS.

Languages

French (fluent)

Spanish (fluent)

English (fluent)

Japanese (Elementary, JLPT
level 4-3)

Github: <https://github.com/sercharpak>

Experienced

Sim4life, Adobe Illustrator, Blender, TensorFlow,
Deeplearning4J, Spark, Scala, Processing, Arduino,
Assembler, UML, MPI, Neuron, Quantum Espresso

References

Prof. Grégoire Courtine

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Prof. Henri Lorach

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Lausanne University (UNIL)
(E-mail) henri.lorach@unil.ch